
Lorenzo Bini

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Education

- University of Geneva**, Ph.D. Candidate - Department of Computer Science & CUI Nov 2022 – Ongoing
- Major: Artificial Intelligence and Machine Learning.
 - Research interests: Graph neural networks, adversarial learning, representation learning, active and self-supervised learning. Generative AI for medicine and healthcare, including 3D genomics, multi-modal learning, and multi-omics/scRNA-seq data generation. Implementation of robust adversarial models within weak/self-supervised training strategies to reduce label acquisition costs.
- Polytechnic of Turin**, Master of Science in Physics of Complex Systems Sep 2020 – Sep 2022
- GPA: 4.0/4.0
- Alma Mater Studiorum - University of Bologna**, Bachelor degree in Physics & Astronomy Systems Sep 2017 – Sep 2020
- GPA: 4.0/4.0

Experience

- PhD Researcher**, [Jian Ma's Lab](#) - Comp. Science Dept., Carnegie Mellon University - Sep 2025 - Ongoing
Joint collaboration with [Harvard University](#) and [Massachusetts Institute of Technology](#)
- Developed and implemented symmetry-aware flow-matching models for 3D genome ensembles operating directly in 3D coordinate space.
 - Built multi-modal conditioning pipelines combining genomic data sources to generate realistic chromosome conformational ensembles.
 - Built multi-modal learning models for multi-omics ensembles generation.
 - Designed and integrated LLM-based multi-modal encoders to fuse epigenomic signals and imaging data.
 - Implemented scalable training & inference (subchain batching, sparse graphs, RMSD alignment) and evaluation tools (contact-map, SCC, insulation score, radius-of-gyration).
 - Optimized memory and runtime (sparse attention) to enable large-region generation.
- PhD Researcher**, Hôpitaux Universitaires de Genève (HUG) – Geneva, Switzerland Nov 2022 - Ongoing
- Detection of minimal residual disease (MRD) of acute lymphoblastic and myeloid leukemia (AML/ALL) from flow cytometry data.
 - Development of fast training/inference deep learning models (e.g, Graph Transformers) for single-cell classification in weak/self-supervised contexts.
 - Development of multi-modal generative models (e.g., DDPMs, Flow Matching) for 3D genomics, flow cytometry, scRNA-seq, spatial transcriptomics, and multiomics data.
 - LLMs to streamline routine hospital processes.
- Teaching Assistant**, University of Geneva – Department of Computer Science & CUI Nov 2022 – Ongoing
- Introduction to Computational Finance - [14X030](#);
 - TALN: Traitement de la langue approches linguistiques et approches empiriques (NLP) - [34C2161](#);
 - Information Retrieval - [14X060](#);
 - Data Science - Analyse et Traitement de l'Information - [14X026](#);
 - Selected Chapters - Game Theory - [14X060](#);
- Research Assistant - Intern**, [Quantum Technology Group](#) - University of Norway Feb 2022 - July 2022
- Worked on quantum theory for entanglement and non-locality in optomechanics continuous variable systems.

- Developed quantum frameworks to analyze two-cavity optomechanics systems.

Publications

ChromWeave: Equivariant Generative Modeling of 3D Genome Organization Shuming Liu*, <i>Lorenzo Bini</i> *, Hongwei Tu, Jian Ma NeurIPS 2025 MLSB Workshop (work done while at Carnegie Mellon University - Broad Institute). Under double-blind review as a journal paper.	Apr 2026
Probing Graph Neural Network Activation Patterns Through Graph Topology Floriano Tori*, <i>Lorenzo Bini</i> *, Marco Sorbi, Stéphane Marchand-Maillet, Vincent Ginis To appear in 2026 (preprint, under double-blind review as a conference paper).	Mar 2026
LapDDPM: A Conditional Graph Diffusion Model for scRNA-seq Generation with Spectral Adversarial Perturbations <i>Lorenzo Bini</i> , Stéphane Marchand-Maillet Conference on Health, Inference, and Learning (CHIL'26), Seattle, WA. ICML'2025 GenBio Workshop: The Second Workshop on Generative AI and Biology, Vancouver.	Jun 2025
Self-Supervised Graph Learning via Spectral Bootstrapping and Laplacian-Based Augmentations <i>Lorenzo Bini</i> , Stéphane Marchand-Maillet Preprint. To appear in 2026, under double-blind review as a journal paper.	May 2025
Massive Activations in Graph Neural Networks: Decoding Attention for Domain-Dependent Interpretability <i>Lorenzo Bini</i> , Marco Sorbi, Stéphane Marchand-Maillet ECAI'2025, Bologna (oral presentation) + ICLR'2025 Workshop XAI4Science: From Understanding Model Behavior to Discovering New Scientific Knowledge, Singapore.	Oct 2024
Injecting Hierarchical Biological Priors into Graph Neural Networks for Flow Cytometry Prediction <i>Lorenzo Bini</i> , Stéphane Marchand-Maillet ICML'2024 + Workshop on Accessible and Efficient Foundation Models for Biological Discovery, Wien, Austria.	Jul 2024
FlowCyt: A Comparative Study of Deep Learning Approaches for Multi-Class Classification in Flow Cytometry Benchmarking <i>Lorenzo Bini</i> , Margarita Liarou, Thomas Matthes, Stéphane Marchand-Maillet Conference on Health, Inference, and Learning (CHIL'24), New-York, NY.	Jun 2024
Why Attention Graphs Are All We Need: Pioneering Hierarchical Classification of Hematologic Cell Populations with LeukoGraph <i>Lorenzo Bini</i> , Thomas Matthes, Stéphane Marchand-Maillet Preprint arXiv:2402.18610, under double-blind review as a conference paper.	Feb 2024
HemaGraph: Breaking Barriers in Hematologic Single Cell Classification with Graph Attention <i>Lorenzo Bini</i> , Thomas Matthes, Stéphane Marchand-Maillet Preprint arXiv:2402.18611, under double-blind review as a conference paper.	Dec 2023

Awards & Oral Presentations

Oral Presentation at the Machine Learning Summer School NYC hosted by Columbia University & Bloomberg - MLSS'26, New York • Invited oral presentation to expose my PhD research at the Machine Learning Summer School NYC hosted by Columbia University in collaboration with Bloomberg (MLSS'26), New York.	Jun. 2026
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Oral Presentation at the Second Workshop on Explainable Artificial Intelligence for the Medical Domain - ECAI'25, Bologna	Oct. 2025
Invited oral presentation of the accepted paper "Massive Activations in Graph Neural Networks: Decoding Attention for Domain-Dependent Interpretability" at European Conference on Artificial Intelligence (ECAI'25), Bologna.	
Winner of Swiss National Science Foundation PhD Mobility Grant	Sep. 2025
Winner of the Swiss National Science Foundation (SNSF) PhD Mobility Grant for a 6-months internship at Ray and Stephanie Lane Computational Biology Department - Carnegie Mellon University (CMU), Pittsburgh USA. Joint collaboration with Harvard University (Broad Institute) and Massachusetts Institute of Technology (MIT EECS).	
PhD Symposium - CHIL'24 at Cornell Tech University, NY	Jun. 2024
Winner of the PhD Symposium money-prize to attend and present PhD work "Adversarial Robust GNNs: Enhancing Learning with Knowledge Injection in Single-Cell Data" at CHIL'24, conference held by Cornell Tech University, New York.	
CHAIR Structured Learning Workshop - Chalmers University of Technology	Oct. 2023
Oral presentation of the "Knowledge Distillation in Acute Myeloid Leukemia Classification: Tabular Data Meets Graph Neural Networks" poster at the AI Structured Learning 2023 Workshop in Göteborg, Sweden.	
Winner of Thesis on Proposal 2021/2022	Feb. 2022
- Winner of the "Thesis on Proposal 2021/2022" call for bids for Master's Thesis on "Entanglement and nonlocality in optomechanics continuous variable systems" under the supervision of Prof. Francesco Pietro Massel & Prof. Vittorio Penna. - Received grants for research period at USN-Kongsberg.	
Projects	
Flow Cytometry Deep Learning Benchmark	FlowCyt-Benchmark
- Developed the first publicly available deep learning benchmark for single cell classification and clustering on flow cytometry data. Tested on a cohort of 30 patients selected by expert hematologists, from bone marrow and peripheral blood samples. Benchmarked multi-modal SOTA classification/generative models including GNNs, GraphTransformers, Diffusion Models (DDPMs), Flow Matching Models and VAEs. - Tools Used: Python, CSS, HTML.	
Hackathons & Competitions	2019 - Ongoing
- Regular participation at Kaggle/LeetCode competitions and Hackathons; runner-up BR41N.IO 2021 Hackathon@PoliTO, runner-up UNIBO-IBM-Unipol Hackathon 2019, 3rd classified QuHack4IA 2023. - Tools Used: Python - Competitive Programming.	
Math/Physics Olympiad	Sep 2014 - Sep 2019
2x winner of the Italian Regional Math Olympiad. 1x winner of the Italian Regional Physics Olympiad.	
Visiting Student , City Montessori School, Lucknow - Uttar Pradesh, India USN	Aug 2016 - Sep 2016
Visited the City Montessori School together with the Italian cultural association "CinemáCinemá" to provide help and needs to elementary/mid school students.	
Oxfam Volunteering	2016 - Ongoing
I do regularly serve as volunteer for charity organization, such as Oxfam Italy.	

Technologies

- **Programming Languages:** Python, Julia, R, , Matlab, Mathematica, C/C++ , CUDA.
- **Frameworks:** Pytorch, Pytorch-Lightning, Tensorflow, Keras, Jupyter.
- **AI/ML:** SciPy, scikit-learn, Github, GitLab, Numpy, Pandas, Matplotlib, Seaborn, wandb, PyG.
- **Tools:** Git, Linux, ssh, SLURM, pip, Anaconda, Docker, VSCode, Bash (Shell).

Software Licenses: Kaluza Analysis Software, Zemax OpticStudio | Comprehensive Optical Design Software, TeXstudio, BioVinci Software.

Reviewing Duties

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| • Conference on Neural Information Processing Systems (NeurIPS). | 2025 - Ongoing |
| • International Conference on Learning Representations (ICLR). | 2023 - Ongoing |
| • International Conference on Machine Learning (ICML). | 2022 - Ongoing |
| • Learning on Graphs Conference (LOG). | 2023 - Ongoing |
| • AAAI Conference on Artificial Intelligence. | 2025 - Ongoing |
| • IEEE Transactions on Computational Biology and Bioinformatics. | 2023 - Ongoing |